

21. The catalytic activity of the transition metals and their compounds is ascribed to their

- (a) Chemical reactivity
- (b) Magnetic behaviour
- (c) Unfilled d-orbitals
- (d) Ability to adopt multiple oxidation states and their complexing ability**

Answer: (d) Ability to adopt multiple oxidation states and their complexing ability

Explanation: Transition metals are effective catalysts because they can change oxidation state and form intermediate complexes/adsorb reactants on their surfaces.

22. What is the general electronic configuration for 2nd row transition series

- (a) $[\text{Ne}]3d^{1-10}4s^2$
- (b) $[\text{Ar}]3d^{1-10}4s^{1-2}$
- (c) $[\text{Kr}]4d^{1-10}5s^{1-2}$**
- (d) $[\text{Xe}]5d^{1-10}5s^{1-2}$

Answer: (c) $[\text{Kr}]4d^{1-10}5s^{1-2}$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

23. Transitional elements are named transition elements because their characters are

- (a) In between s and p - block elements**
- (b) Like that of p and d - block elements
- (c) They are members of I – A group
- (d) They are like inactive elements

Answer: (a) In between s and p - block elements

Explanation: The correct choice is In between s and p - block elements. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

24. Those elements whose two outermost orbitals are incompletely filled with electrons are

- (a) p - block elements
- (b) s - block elements
- (c) Transitional elements**
- (d) Both s and p - block elements

Answer: (c) Transitional elements

Explanation: The correct choice is Transitional elements. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

25. Which ion has maximum magnetic moment

- (a) V^{+3}
- (b) Mn^{+3}
- (c) Fe^{+3}**
- (d) Cu^{+2}

Answer: (c) Fe^{+3}



Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

26. Which of the following transition metal is present in misch metal

- (a) La
- (b) Sc
- (c) Ni
- (d) Cr

Answer: (a) La

Explanation: Misch metal is mainly a mixture of lanthanoid (rare-earth) metals, commonly with a small amount of iron.

27. Which of the following statements is not true in regard to transition elements

- (a) They readily form complex compounds
- (b) They show variable valency
- (c) All their ions are colourless
- (d) Their ions contain partially filled d - electron levels

Answer: (c) All their ions are colourless

Explanation: The correct choice is All their ions are colourless. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

28. Which of the following represents the electronic configuration of a transition element

- (a) $1s^2, 2s^2p^6 \dots ns^2p^3$
- (b) $1s^2, 2s^2p^6 \dots ns^2p^6d^3, (n+1)s^2$
- (c) $1s^2, 2s^2p^6 \dots ns^2p^6d^{10}, (n+1)s^2p^1$
- (d) $1s^2, 2s^2p^6 \dots ns^2p^6$

Answer: (b)

$1s^2, 2s^2p^6 \dots ns^2p^6d^3, (n+1)s^2$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

29. The general electronic configuration of transition elements is

- (a) $(n-1)d^{1-5}$
- (b) $(n-1)d^{1-10}ns^1$
- (c) $(n-1)d^{1-10}ns^{1-2}$
- (d) $ns^2(n-1)d^{10}$

Answer: (c) $(n-1)d^{1-10}ns^{1-2}$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

30. Transition elements are coloured

- (a) Due to small size
- (b) Due to metallic nature
- (c) Due to unpaired d - electrons
- (d) All of these



Answer: (c) Due to unpaired d - electrons

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d^0 or d^{10} ions are generally colourless unless charge-transfer effects operate.

31. Which of the following has the maximum number of unpaired d-electrons

- (a) Zn
- (b) Fe^{2+}**
- (c) Ni^{3+}
- (d) Cu^+

Answer: (b) Fe^{2+}

Explanation: The correct choice is Fe^{2+} . This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

32. Which is not amphoteric

- (a) Al^{3+}
- (b) Cr^{3+}
- (c) Fe^{3+}**
- (d) Zn^{2+}

Answer: (c) Fe^{3+}

Explanation: The correct choice is Fe^{3+} . This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

33. Which does not form amalgam

- (a) Fe**
- (b) Co
- (c) Ag
- (d) Zn

Answer: (a) Fe

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.

34. Transition metals are often paramagnetic owing to

- (a) Their high M.P. and B.P.
- (b) The presence of vacant orbitals
- (c) The presence of one or more unpaired electrons in the system**
- (d) Their being less electropositive than the elements of groups I-A and II-A

Answer: (c) The presence of one or more unpaired electrons in the system

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d^0 and d^{10} species are diamagnetic.

35. Elements which generally exhibit multiple oxidation states and whose ions are usually coloured

- (a) Metalloids
- (b) Transition elements**
- (c) Non-metals
- (d) Gases



Answer: (b) Transition elements

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

36. Which of the following transition metal cation has maximum unpaired electrons

- (a) Mn^{+2}
- (b) Fe^{+2}
- (c) Co^{+2}
- (d) Ni^{+2}

Answer: (a) Mn^{+2}

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

37. Maximum number of oxidation states of transition metal is derived from the following configuration

- (a) ns electron
- (b) (n-1)d electron
- (c) (n+1)d electron
- (d) $ns+(n-1)d$ electron

Answer: (d) $ns+(n-1)d$ electron

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

38. Which of the following statement is correct

- (a) Iron belongs to 3rd transition series of the periodic table
- (b) Iron belongs to f-block of the periodic table
- (c) Iron belongs to second transition series of the periodic table
- (d) Iron belongs to group VIII of the periodic table

Answer: (d) Iron belongs to group VIII of the periodic table

Explanation: The correct choice is Iron belongs to group VIII of the periodic table. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

39. Zinc does not show variable valency like d-block elements because]

- (a) It is a soft metal
- (b) d-orbital is complete
- (c) It is low melting
- (d) Two electrons are present in the outermost orbit

Answer: (b) d-orbital is complete

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.



40. Which of the following is a transitional element

- (a) Al
- (b) As
- (c) Ni
- (d) Rb

Answer: (c) Ni

Explanation: The correct choice is Ni.

This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

